

# Sophia Thompson

Irvine, CA | sthom954@gmail.com | +1 (707) 800 5033  
LinkedIn: sm-thompson | GitHub: smthom1

## SUMMARY

Master of Data Science student with a strong foundation in **data analytics, machine learning, and software development**. Experienced in applying computational tools to **environmental, social, and healthcare challenges**, bridging technical innovation with human-centered understanding. Passionate about leveraging **data-driven insights for sustainable, equitable, and impactful solutions**.

## EDUCATION

|                                                                                                                                             |                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <i>University of California, Irvine</i><br><b>Master of Data Science</b>                                                                    | Irvine, CA   <i>(expected) Dec 2026</i> |
| <i>University of California, Davis</i><br><b>B.S. in Computer Science</b><br>Minors: Sociology, Environmental Horticulture & Urban Forestry | Davis, CA   <i>June 2025</i>            |

## SKILLS

|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b> Python, SQL, Java, JavaScript, C, C++, R, LaTeX                                               |
| <b>Data Science:</b> Machine Learning, AI, Statistics, Data Visualization, Predictive Modeling                    |
| <b>Tools &amp; Databases:</b> MySQL, MongoDB, Firebase, PostgreSQL, Streamlit, Git, JSON, Figma                   |
| <b>Concepts:</b> Environmental Informatics, Sustainable Systems, Agent-Based Modeling, Human-Computer Interaction |

## PROFESSIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Research Assistant – AI Lab, UC Irvine</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Irvine, CA   Jan 2026 – Mar 2026 |
| <ul style="list-style-type: none"><li>Engineered R pipelines (dplyr, stringr) to <b>clean and merge large-scale datasets</b>, resolving complex text anomalies for downstream economic modeling.</li><li>Quantified technology adoption and market demand by <b>aggregating multi-source usage statistics and repository metrics</b> across diverse programming languages, identifying macro-level ecosystem trends.</li><li><b>Outcome:</b> Delivered unified statistical summaries of software ecosystem usage, directly enabling the research team to build accurate economic models.</li></ul> |                                  |
| <b>Undergraduate Research Assistant – Urban Science Lab, UC Davis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Davis, CA   Jan 2025 – July 2025 |
| <ul style="list-style-type: none"><li>Collaborated with the <b>Foundation for Food and Agriculture Research (FFAR)</b> to build the <b>Edible Trees Database</b>, improving access to agroforestry resources.</li><li>Developed maintenance programs using <b>Python, R, and SQL</b> to increase data reliability and scalability.</li><li><b>Outcome:</b> Streamlined data workflows, enhancing accessibility for urban forestry researchers.</li></ul>                                                                                                                                           |                                  |

### Data Analyst Intern – *EcoTelesis*

Agoura Hills, CA (remote) | Oct 2024 – Mar 2025

- Analyzed **solid waste and greenhouse gas emissions** data for Los Angeles County, contributing to sustainability efforts.
- Applied **data mining and statistical analysis** to evaluate recycling efficiencies and waste recovery initiatives.
- **Outcome:** Delivered actionable insights supporting county-wide sustainability policies.

### Undergraduate Research Assistant – *Tree Systems Lab, UC Davis*

Davis, CA | Mar 2024 – Sept 2024

- Conducted research in **plant physiology and agro-ecological systems** using Python and R.
- Led development of **machine learning tools** for pistachio image recognition with CVAT and YOLO.
- **Outcome:** Improved agricultural precision and reduced data processing time.

## PROJECTS

---

### Health & Accessibility Platforms – *HackDavis*

*Hackathon Wins: Best Medical Hack (HAPPI, 2025) & Best Use of MongoDB (IntelliConverse, 2023)*

- **HAPPI:** Developed a digital platform to assist Alzheimer's and dementia patients through digitized **SAGE cognitive exams**, using **Streamlit, Firebase, MongoDB Atlas**, and the **Google Gemini API** for personalized cognitive and mobility activity recommendations.
- **IntelliConverse:** Built an accessibility-focused **learning assistance app** with **React, Next.js, Azure Speech, Milvus, and MongoDB**, improving document comprehension for individuals with learning differences.
- **Impact:** Recognized for innovation, inclusivity, and technical execution in healthcare and accessibility-focused hackathons.

### Pistachio Image Recognition (Lead Developer)

*Tree Systems Lab, UC Davis* | Apr 2024 – Sept 2024

- Built **Python-based machine learning models** for automated pistachio classification and sorting.
- Designed image processing filters for background removal and improved detection accuracy.
- **Result:** Reduced sorting errors and material waste, improving agricultural efficiency.

### Edible Trees Database

*Urban Science Lab, UC Davis* | Jan 2025 – July 2025

- Designed a **PostgreSQL/PostGIS database** cataloging edible urban tree species.
- Integrated **Leaflet and Java visualizations** for proof-of-concept web design.
- **Result:** Created scalable infrastructure for public urban forestry data access.

### Agent-Based Model of Infection Dynamics

*UC Davis (PSC 120)* | Sept 2024 – Dec 2024

- Built a **Java-based MASONplus9 simulation** to model SIR-based disease spread dynamics.
- Created interactive GUI for variable adjustment and visualization.
- **Insight:** Supported analysis of epidemic behavior and public health response modeling.